

Appendix: A fragile guardrail—The minimal effect of election fraud on protest intensity in non-democracies

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1 Summary statistics

Table [1](#) provides summary statistics for the variables used in the main text.

Table 1: Summary statistics and variable distributions

	Unique (#)	Missing (%)	Mean	SD	Min	Median	Max
Judicial independence	257	0	0.4	0.2	0.0	0.4	0.9
Legislative constraints	247	10	0.4	0.2	0.0	0.3	0.9
Alternative information	313	0	0.5	0.2	0.0	0.6	0.9
Liberal democracy index	230	0	0.2	0.1	0.0	0.2	0.5
GDP growth rate	250	2	0.0	0.4	-0.8	0.0	5.9
Civil society openness	266	0	0.4	1.0	-2.7	0.4	2.5
National party organization	236	0	0.4	0.9	-2.3	0.5	3.3
Election fraud	386	1	1.0	1.0	-2.3	1.0	3.2
Presidential election	2	0	1.4	0.5	1.0	1.0	2.0
Incumbent term-limited	3	2	0.1	0.2	0.0	0.0	1.0
Physical integrity index	278	0	0.5	0.2	0.0	0.4	0.9
Urbanization	426	0	45.4	19.0	6.8	43.1	100.0
Winning vote-share	424	9	0.6	0.2	0.1	0.5	1.0
Number of pre-election protests	32	0	2.3	9.0	0.0	0.0	100.0
Off-schedule election	2	0	1.5	0.5	1.0	1.0	2.0
Regional protest diffusion	2	0	0.2	0.4	0.0	0.0	1.0
Privatization	246	0	0.1	1.1	-3.9	0.2	2.3
Professional bureaucracy	165	5	-0.3	0.9	-2.7	-0.4	3.3
EMB autonomy	286	0	-0.2	1.1	-2.6	-0.2	2.7

2 Robustness check: Two alternative measures of election manipulation

To ensure that the results presented in the main text are not idiosyncratic to the chosen measure of election fraud, this section replaces that measure with two alternative indicators of election integrity. The first variable is ‘v2xel_frfair,’ from V-Dem, which is a comprehensive indicator of quality of the electoral process, including measures of electoral management body autonomy and capacity, vote-buying, intimidation, fraud, and an overall assessment of the fairness of the election. This is a retroactively coded, expert assessment (as is the variable in the main text), but captures a wider array of possible abuses of electoral integrity.

The second measure reflects contemporaneous judgments by election observers, as reported in the Dataset on International Election Monitoring, or DIEM (Kelley 2011). I use the variable “Explicit cheating,” as a parallel to the V-Dem “intentional irregularities” variable from the main text. This variable is categorical, and so captures less variation than the original measure. However, it helps address the concern that the findings could be systematically biased by the human coders utilized in the V-Dem project (Little and Meng 2024). While there is some evidence that such bias is small (Weidmann 2023), it is still useful to check the V-Dem coders’ assessments against election monitors’ reports.

Both variables produce the same substantive results in the Cox proportional hazards models of protest duration; more election manipulation is associated with shorter protests.

2.1 Overall election integrity

The ‘election integrity’ variable from V-Dem (v2xel_frefair) ranges from 0 to 1, with higher values indicating freer and fairer elections. Note that the sign of this variable is the inverse of the variable used in the main text. The models presented below utilize this variable in place of the main measure of election fraud, while retaining all other control and explanatory variables.

The same interaction relationship between overall election integrity, incumbent vote-share, and protest as is shown for fraud in the main text. Table 2 shows that cleaner elections are associated with a slower end to protest movements. When elections are more unfair overall, incumbents are able to more quickly bring protest to an end. Figure 1 shows that this effect is significant in interaction with low-to-moderate incumbent vote-shares.

2.2 Election observers' reports of fraud

This dataset codes the judgments of international election monitoring missions as categorical variables. The explicit cheating variable is categorical, ranging from the 0 for the cleanest elections to 3 for the most fraudulent. While this data is closely tied to the facts on the ground, as reported by election monitors, it covers a much smaller range of cases than V-Dem: of the 661 observations in the full dataset, 471 do not have DIEM values. This is, in part, due to differences in the time periods covered (DIEM ranges from 1980 to 2004, while the main dataset covers 1989 - 2012). This problem is compounded by the fact that most elections in the dataset go un-protested. Of the 156 cases where protest occurs, 110 lack data in DIEM. While this limits the usefulness of the DIEM variable, it can still provide some confirmatory information for the main analysis.

Table 2: Cox proportional hazard models of protest duration

	Model 1	Model 2
Leg. constraints (lag)	−0.899 (0.624)	−0.863 (0.621)
Alternative info. (lag)	2.193* (1.002)	2.391* (1.018)
Civil soc. openness (lag)	−0.421* (0.189)	−0.434* (0.189)
GDP growth rate (lag)	0.197 (0.184)	0.197 (0.186)
National party organization (lag)	−0.125 (0.130)	−0.137 (0.133)
Incumbent vote-share	−0.319 (0.599)	−1.869 (1.350)
Presidential election	−0.159 (0.218)	−0.174 (0.218)
Incumbent term-limited	−0.419 (0.337)	−0.485 (0.340)
Phys. integrity index (lag)	2.695 (2.217)	2.435 (2.225)
Phys. int. lag squared	−2.126 (2.374)	−1.812 (2.387)
Urbanization	0.002 (0.006)	0.003 (0.006)
Rentier state	−0.235 (0.323)	−0.285 (0.330)
Tax state	0.008 (0.302)	−0.037 (0.308)
Off-schedule election	0.469* (0.229)	0.440+ (0.229)
Regional protest diffusion	−0.017 (0.221)	−0.002 (0.221)
Overall election integrity	−2.423** (0.875)	−5.961* (2.904)
N. post-election protest	−0.053*** (0.013)	−0.053*** (0.013)
Overall election integrity:Incumbent vote-share		6.190 (4.823)
Num.Obs.	130	130
RMSE	0.92	0.93

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

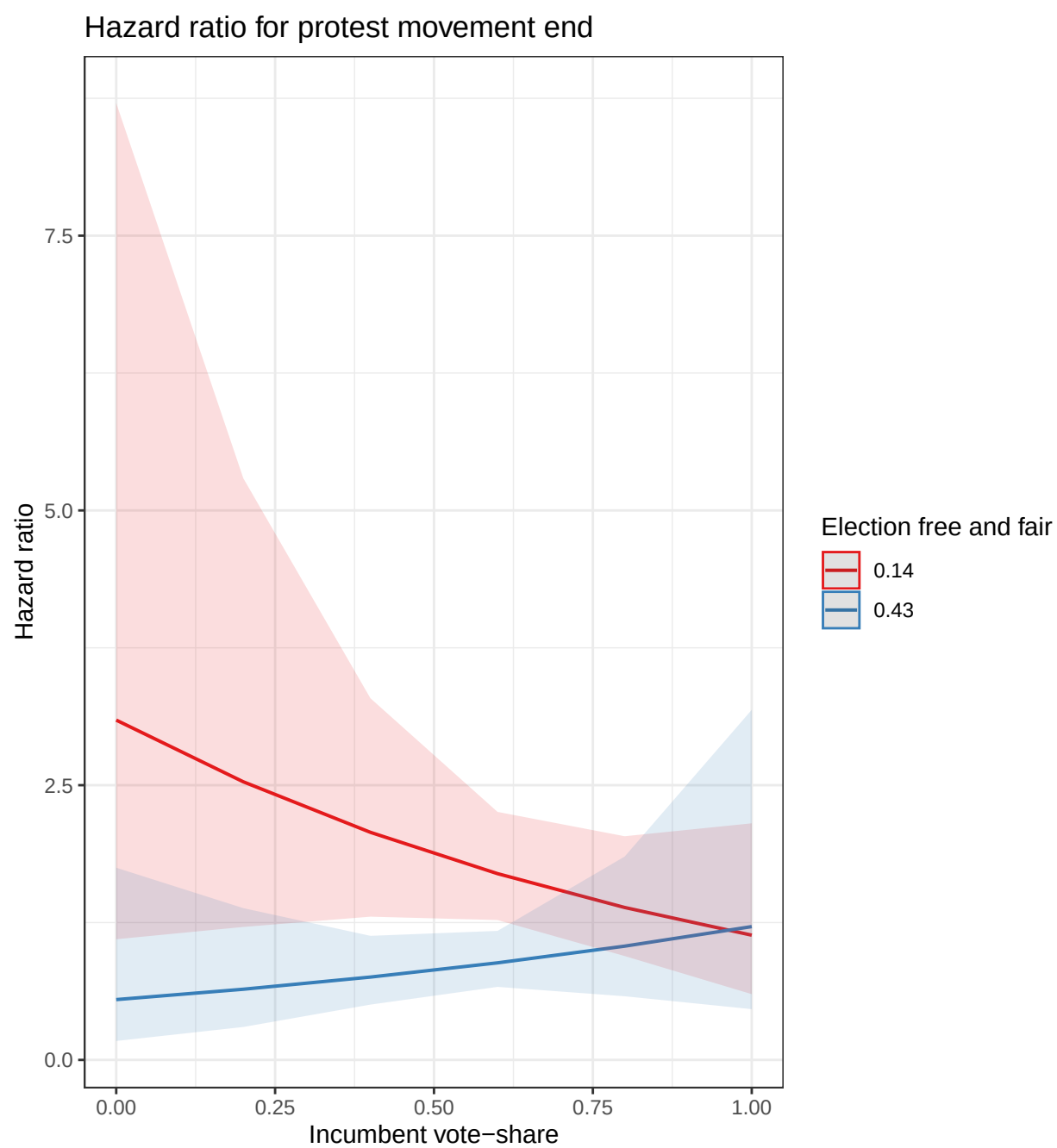


Figure 1: Fraud, winning vote-share, and duration of protest

Table 3: Models using DIEM measure of election fraud

	Model 3	Model 4
Leg. constraints (lag)	1.233 (1.435)	1.241 (1.441)
Alternative info. (lag)	0.640 (2.509)	0.634 (2.518)
Civil soc. openness (lag)	-0.486 (0.407)	-0.483 (0.409)
GDP growth rate (lag)	0.402* (0.198)	0.400* (0.199)
National party organization (lag)	-0.298 (0.261)	-0.301 (0.265)
Incumbent vote-share	-1.471 (1.188)	-1.119 (3.747)
Presidential election	-0.353 (0.470)	-0.359 (0.473)
Incumbent term-limited	0.333 (0.839)	0.341 (0.843)
Phys. integrity index (lag)	4.334 (5.421)	4.243 (5.498)
Phys. int. lag squared	-4.258 (5.863)	-4.196 (5.898)
Urbanization	0.003 (0.014)	0.004 (0.014)
Off-schedule election	0.561 (0.559)	0.547 (0.576)
Regional protest diffusion	-0.466 (0.428)	-0.474 (0.435)
Explicit cheating (DIEM)	0.582* (0.280)	0.655 (0.788)
N. post-election protests	-0.059* (0.023)	-0.060* (0.024)
Incumbent vote-share:Explicit cheating (DIEM)		-0.143 (1.448)
Num.Obs.	44	44
AIC	251.1	253.1
BIC	277.9	281.7
RMSE	0.85	0.85

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3 reports two model where the DIEM measure as the explanatory variable in a

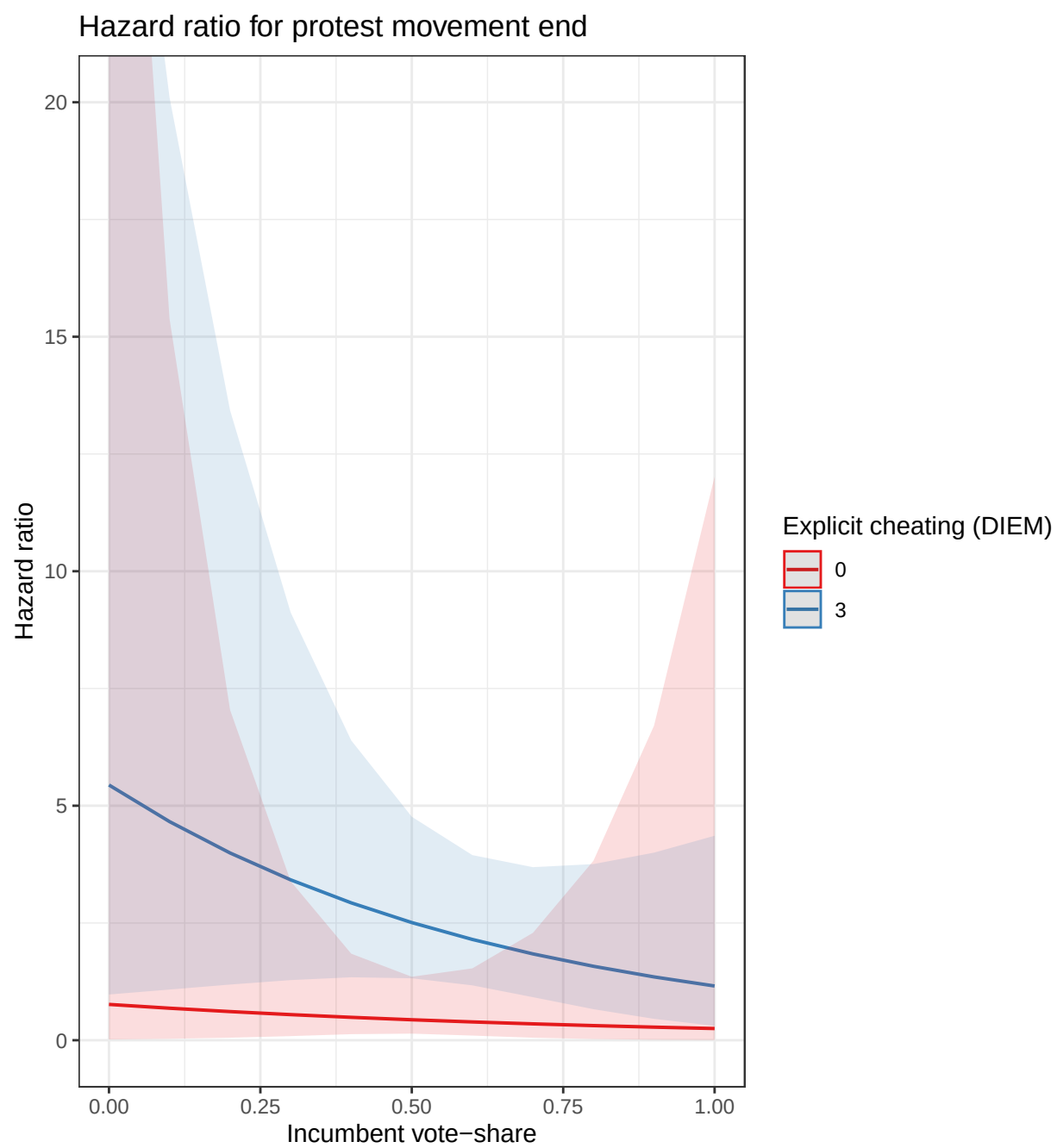


Figure 2: Fraud, winning vote-share, and duration of protest (DIEM)

survival model of protest movements, both as standalone variable and in interaction with the incumbent’s vote-share. Here the difficulty with missingness becomes apparent, with only 44 observations in the models. Figure 2 shows that higher ‘explicit cheating’ has a negative effect on protest duration around the fifty-percent vote-share mark, though the effect is small and falls just short of conventional significance. Given the small number of observations, it is even more reassuring that the expected relationships between fraud and a quick end to protest movements are detected.

3 Addressing concerns about selection effects

3.1 Predicted and residual levels of fraud

The main text focuses on elections where the incumbent claims victory. However, classic regime-weakness models also predict that some incumbents who will voluntarily avoid electoral manipulation and accept a loss at the ballot box. If this is the case, it could introduce selection bias into the sample for the duration analyses in the main paper by sorting only the strongest incumbents into the sample that faces protest.

Fully investigating this possibility should be the focus of future work, but some initial analyses are provided here to show that this is not the case. The analysis in this subsection proceeds in three steps. First, using all the observations in the dataset—winning and losing incumbents—I estimate the predicted level of election fraud for each case, using only structural variables (i.e. professionalization of the bureaucracy, electoral management body autonomy, urbanization, etc) in an OLS model. Factors associated with deterrence, like the number of pre-election protests are excluded. This should produce an estimate of the level of manipulation the incumbent could produce in the absence of strategic deterrence.

Second, the residuals from this first model are then used as a dependent variable in second model. The residuals reflect the degree to which an incumbent over- or under-produced manipulation, relative to their expected baseline. If the deterrence interpretation is correct (which may lead to selection effects), the indicators of protest risk should be negatively associated with these residuals. In other words, higher protest risk should lead incumbents to under-produce manipulation. Under the regime-strength logic, we might expect the opposite, as incumbents work hard to build and indicate strength in riskier environments. This model is also an OLS model.

Lastly, two logit models of incumbent loss are presented, using the fitted values from the first model and indicators of protest risk as predictors. The fitted values indicated the expected, structural level of electoral manipulation. Here, the deterrence logic would expect that indicators of pre-election protest risk will make incumbents more likely to lose, while the regime-strength model would expect that incumbent loss will be largely driven by structural factors.

Table 4 shows the results. Model 5 shows how structural factors are predictors of the level of observed manipulation. We observe some expected results; for instance, more autonomous EMBs and more professionalized bureaucracies are associated with lower fraud. Model 6

shows that indicators of protest risk—the number of pre-election protests, and the existence of a recent successful protest movement in the region—are not associated with lower fraud. In fact, just the opposite: incumbents appear to manipulate more heavily under riskier conditions. This is a key finding, and is evidence against the classic deterrent approach; it also suggests that weaker governments are not voluntarily dialing back manipulation to avoid protest, mitigating concerns about selection effects in the main models.

Models 7 and 8 take the fitted values from the first model and employ them as an explanatory variable for incumbent loss, while controlling for pre-election protest. As both models show, the more incumbents invest in the structural prerequisites of manipulation, the less likely they are to lose elections. By contrast, we observe little or no effect for pre-election protest risk. Model 7 shows no effect for regional protest diffusion on incumbent loss, while Model 8 shows a significant ($p = 0.05$) but substantively small effect for pre-election protest.

All told, this analysis supports the argument in the main text, which holds that the capacity to manipulate heavily is associated with incumbent durability. The results in this section are consistent with a world in which losing incumbents are not voluntarily surrendering power in the face of a committed opposition (on average), but are failing to produce sufficient election manipulation to win due to structural constraints. This should reduce the concern that the main findings are driven by an endogenous process, in which incumbents who face more organized and capable opposition actors are strategically avoiding protest by producing less fraud.

Table 4: Models of manipulation and residuals (OLS) and incumbent loss (logit)

	Model 5	Model 6	Model 7	Model 8
	Fraud	Fraud residual	Inc. loss	Inc. loss
(Intercept)	1.031*** (0.177)	−0.045 (0.033)	−0.262 (0.167)	−0.318+ (0.167)
Judical independence (lag)	−1.206*** (0.192)			
GDP growth rate (lag)	0.175* (0.084)			
Presidential election	−0.054 (0.061)			
Urbanization	−0.008*** (0.002)			
Rentier state	0.220** (0.082)			
Tax state	0.209* (0.081)			
Phys. integrity index (lag)	2.860*** (0.568)			
Phys. int. lag squared	−3.353*** (0.588)			
Private ownership (lag)	0.191*** (0.031)			
Professional bureaucracy (lag)	−0.168*** (0.039)			
EMB autonomy	−0.394*** (0.036)			
Number of pre-election protests		0.006* (0.003)		0.019+ (0.010)
Regional protest diffusion		0.183* (0.079)	−0.159 (0.337)	
Expected fraud level			−1.151*** (0.168)	−1.182*** (0.167)
Num.Obs.	605	605	529	529
R2	0.482	0.016		
R2 Adj.	0.472	0.013		
AIC	1349.9	1321.9	506.7	503.3
BIC	1407.2	1339.5	519.5	516.1
F	50.138	5.035		

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

3.2 Incumbent defeat by protest

A second possible source of selection bias in the main analysis sample is the decision to exclude cases where incumbents were defeated by protest movements. While this decision is necessary given the research design, it might still introduce bias if—for instance—many defeated incumbents manipulated heavily. The main text does not examine successful protests, which are a small minority of cases, but I include them in an analysis here. There is little to suggest that incumbents who manipulate more are more vulnerable to defeat by protesters.

First, Figure 3 shows cases of ‘successful’ and unsuccessful protest at varying levels of election fraud. Protests are coded as successful if they lead to incumbent ouster or to new elections as recorded in NELDA. As the figure shows, the large majority of protests are unsuccessful by this measure. Moreover, the small number of successful cases appear spread over approximately the same range of election fraud as unsuccessful cases; we do not observe a clustering of successful protest at higher levels of fraud.

Next, Table 5 reports three logit regression models of successful protest; the dataset for these models includes all cases with a least one protest event. Model 9 is a minimal model that includes *intentional election irregularities*, *incumbent vote-share* and a control for *presidential elections*. Model 10 includes a larger number of control variables, while Model 11 adds the interaction term. As the results in the table show, greater election manipulation is not associated with a higher likelihood of incumbent defeat in any model. Figure 4 shows no significant effect of election manipulation on protest success across all values of incumbent vote-share. Taken together, this is further evidence against the fraud-as-weakness model, and suggests the main results are not biased by excluding high-fraud cases where protesters toppled the government.

4 Removing controls to address multicollinearity

The models in the main text include a suite of control variables aimed at controlling for common causes of election manipulation and protest, in order to minimize the risks of confounding and omitted variable bias. However, this approach runs the risk that individual coefficients may be highly sensitive to model specification due to multicollinearity. In particular, many of the control variables—such as *legislative constraints*, *judicial independence*, and *physical integrity* are likely to be correlated with pre-election protest. To address this risk, the models in Table 6 exclude non-electoral control variables. That is, they include variables related to the election itself, the measure of *pre-election protest*, and the explanatory variables from the main models.

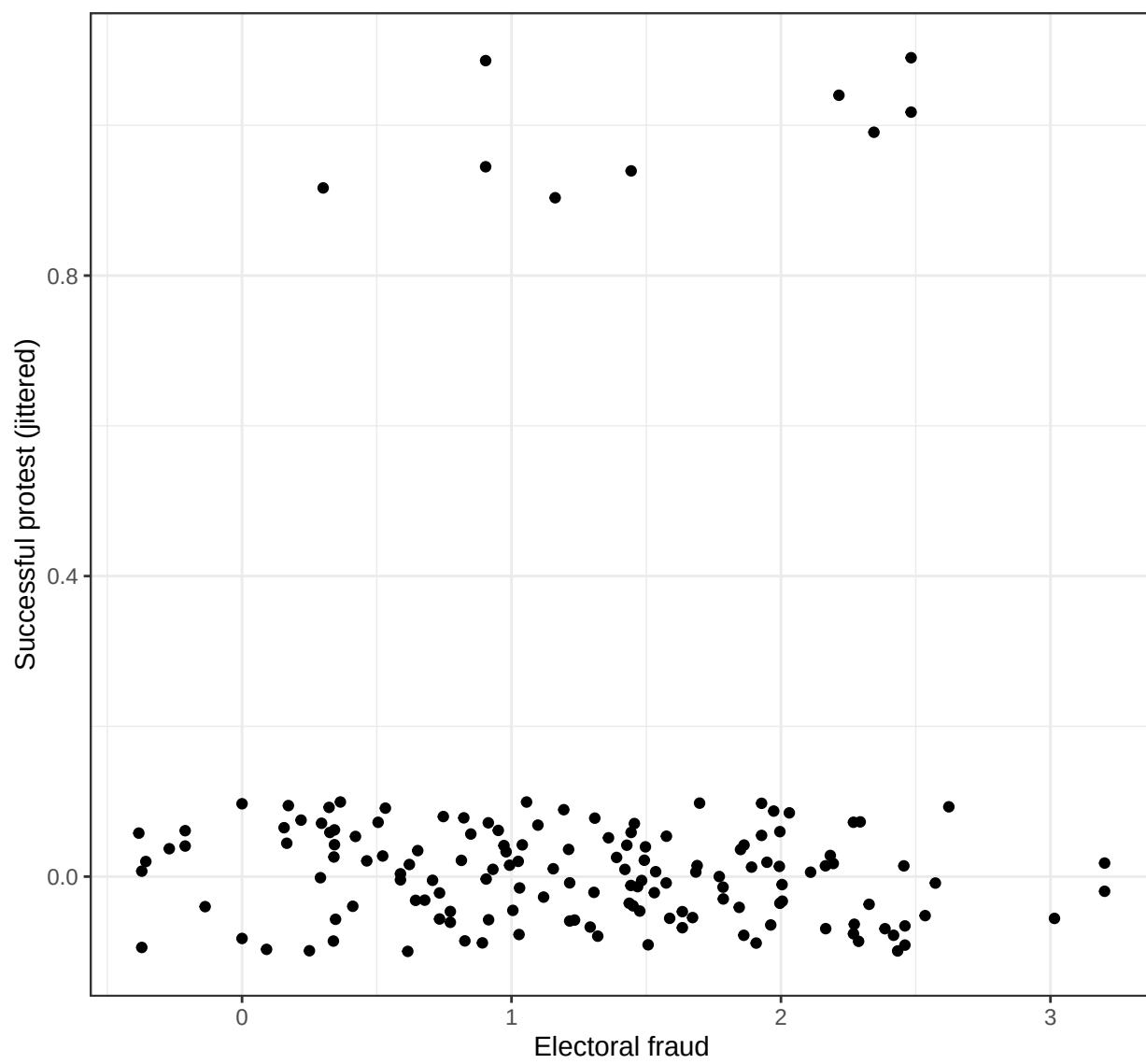


Figure 3: Election fraud and successful protest

Table 5: Logit models of incumbent ouster by protest

	Model 9	Model 10	Model 11
(Intercept)	−4.167 (2.620)	−7.157+ (3.887)	−8.340+ (4.430)
Presidential election	0.698 (0.880)	−1.131 (1.466)	−1.226 (1.500)
Election fraud	1.538 (1.278)	0.153 (0.758)	1.015 (1.698)
Winner vote-share	0.293 (4.333)	0.206 (2.941)	3.162 (5.917)
Election fraud:Winner vote-share	−1.915 (2.236)		
Civil soc. openness (lag)		0.533 (0.673)	0.497 (0.664)
GDP growth rate (lag)		−6.036 (6.490)	−6.125 (6.622)
Incumbent term-limited		2.803+ (1.658)	2.893+ (1.697)
Phys. integrity index (lag)		0.737 (9.932)	−0.572 (10.227)
Phys. int. lag squared		−4.805 (11.381)	−3.480 (11.591)
Urbanization		0.056+ (0.033)	0.056+ (0.033)
Off-schedule election		−0.634 (1.440)	−0.809 (1.499)
Regional protest diffusion		3.542** (1.375)	3.534* (1.384)
Winner vote-share:Election fraud			−1.687 (2.937)
Num.Obs.	143	143	143

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

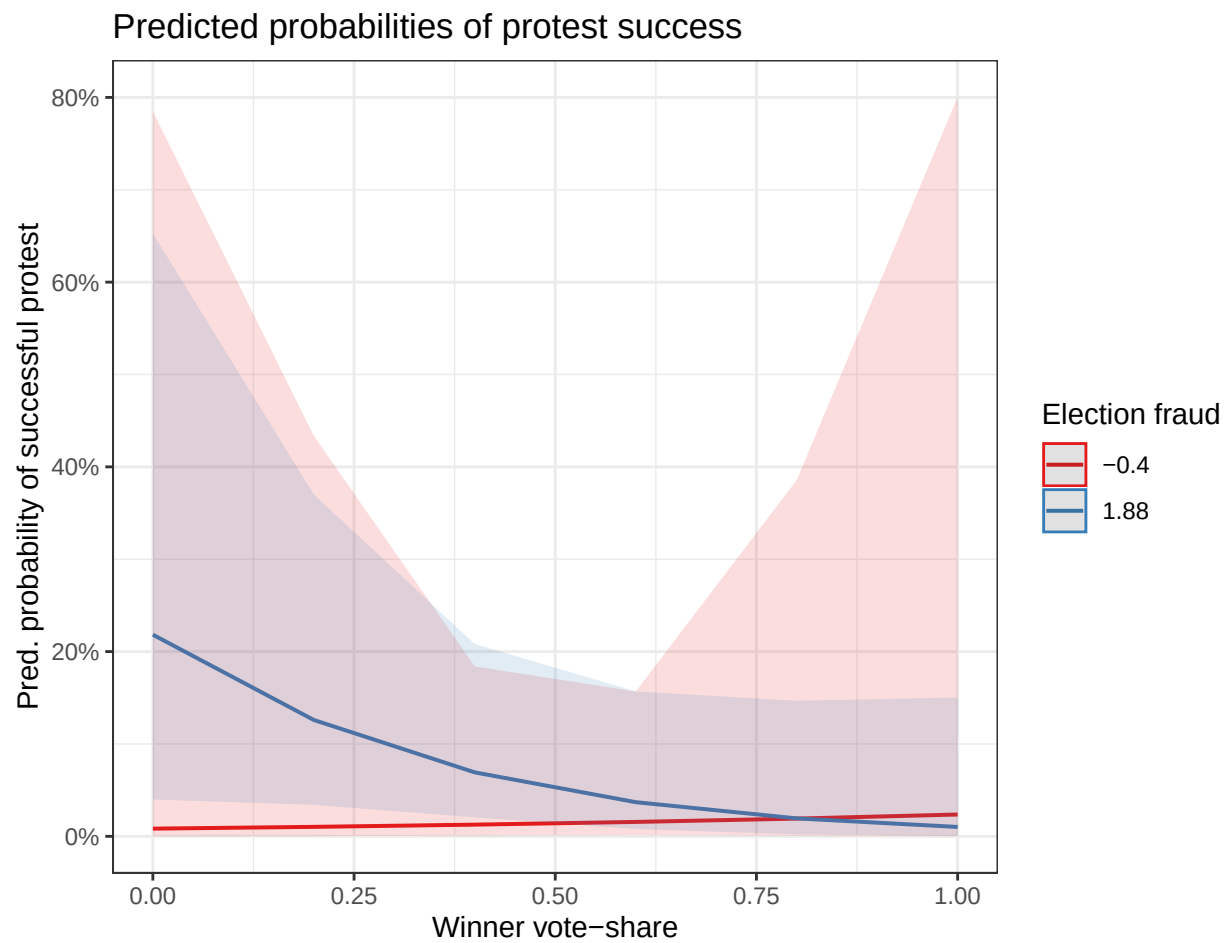


Figure 4: Fraud, winning vote-share, and incumbent ouster

Table 6: Cox proportional hazard models of protest duration

	Model 12	Model 13
Incumbent vote-share	−0.138 (0.523)	1.190 (0.897)
Presidential election	−0.134 (0.188)	−0.139 (0.190)
Incumbent term-limited	−0.243 (0.309)	−0.266 (0.308)
Off-schedule election	0.300 (0.199)	0.262 (0.203)
Regional protest diffusion	−0.086 (0.201)	−0.155 (0.204)
Election fraud	0.391** (0.120)	0.933** (0.324)
N. post-election protests	−0.048*** (0.011)	−0.048*** (0.011)
Election fraud:Incumbent vote-share		−0.903+ (0.505)
Num.Obs.	141	141
RMSE	0.98	0.98

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The results echo those from the main text. Fraudulent elections are associated with a higher likelihood that protest movements will fail quickly, as Figure 5 shows.

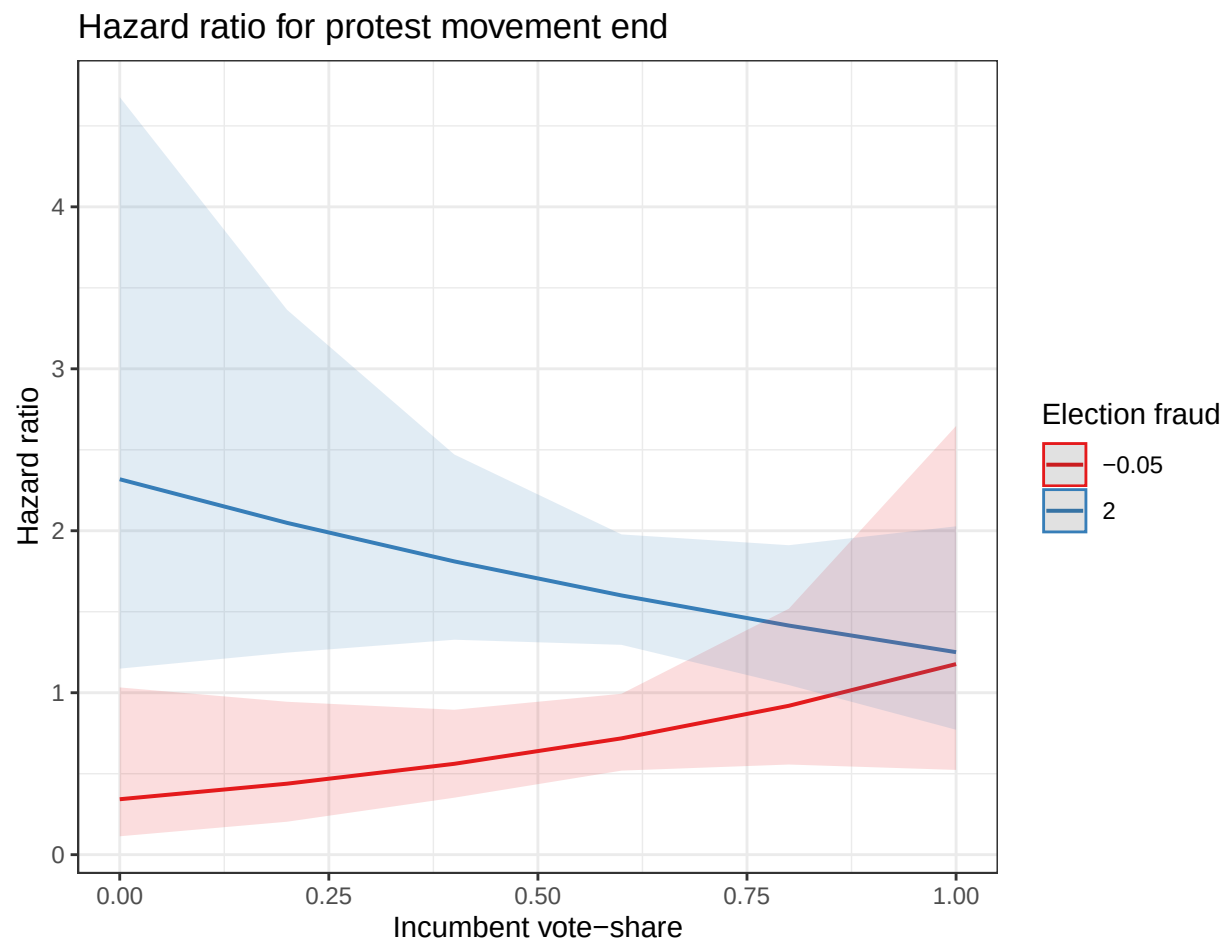


Figure 5: Fraud, winning vote-share, and duration of protest

5 Confirming assumptions: Fraud and the likelihood of protest

Both the regime-weakness and regime-strength models of manipulation predict that protest initiation should be more likely for incumbents with low vote-share and high levels of manipulation, compared to incumbents with low vote-share and low levels of manipulation. In the regime-weakness model, this dynamic reflects the very low genuine popularity of the former type of incumbent. For the regime-strength model, it reflects high grievances and a high likelihood that the ‘stolen’ election will become a coordination device for protesters. Using the same ECAV data from the main text, I now model protest initiation as a binary variable that takes on a value of 1 if any electoral protests occurred, and zero otherwise. Since the dependent variable is binary, it is modeled using standard logistic regression.

Table 7 shows the results. Model 14 is a base model, where incumbent loss, winner vote-share, and *election fraud* are standalone variables. In keeping with the general (though inconsistent) findings in the prior literature, there is a significant positive relationship between fraud severity and the likelihood of protest initiation. The winner’s claimed margin of victory is negatively associated with protest initiation. Model 15 includes the three-way interaction between *election fraud*, incumbent loss, and winning vote-share, with no controls other than a dummy variable for *presidential elections*. This allows for the largest possible sample size, and shows that the coefficient signs are consistent with those of the full Model 16.

Given the interaction term in Model 16, Figure 6 illustrates the effects of election fraud on protest risk across various levels of winning vote-share and incumbent loss¹. The values for election fraud represent one standard deviation above and below the mean value, or about the 15th and 85th percentiles, respectively.

The direction of the effect supports both theories. When an incumbent claims victory (right panel) and their vote-share is low, electoral protest is significantly likelier when fraud is elevated than when the election is relatively clean. Furthermore, protest against winning incumbents is significantly likelier when fraud is high and vote-shares are low, compared to an election with an equivalent level of fraud but high claimed vote-shares. As noted above, such a finding could be consistent with either the regime-weakness or regime-strength models of manipulation.

6 Graphical presentation of protest duration

While the regression models in the main text allow for the use of control variables, it is also possible to visually examine the pattern of protest duration with regard to incumbent vote-share and degree of election fraud. Since the regime-strength model makes predictions based on joint values of incumbent vote-share and election fraud, a standard X-Y scatterplot is not sufficient. Instead, the data must be examined in three dimensions; Figure 7 does so by presenting a contour plot created using the `latticeExtra` package in R. In the figure, darker areas indicate conceptual regions where incumbents are weaker. Each dot represents a

¹All other variables constant at their means (continuous variables) or modes (categorical variables).

Table 7: Table 3: Logit models of protest occurrence

	Model 14	Model 15	Model 16
(Intercept)	−0.272 (0.308)	−1.747** (0.622)	−6.597*** (1.677)
Incumbent loss	−0.266 (0.254)	1.019 (0.874)	4.448** (1.633)
Election fraud	0.247* (0.101)	1.333** (0.421)	2.883*** (0.750)
Winner vote-share	−1.046* (0.468)	0.639 (0.986)	4.736** (1.733)
Winner vote-share:Election fraud		−1.765** (0.634)	−4.605*** (1.154)
Winner vote-share:Inc. loss		−2.385 (1.726)	−10.071** (3.772)
Election fraud:Inc. loss		−1.094 (0.698)	−4.161*** (1.188)
Judicial independence (lag)			2.762+ (1.615)
Leg. constraints (lag)			0.197 (0.931)
GDP growth rate (lag)			0.063 (0.538)
Alternative info. (lag)			−0.602 (1.475)
Civil soc. openness (lag)			0.309 (0.261)
Incumbent term-limited			−0.961 (0.729)
Urbanization			0.006 (0.010)
Presidential		1.000*** (0.205)	1.866*** (0.393)
Off-schedule election			−0.308 (0.359)
Number of pre-election protests			21.326 (1145.747)
Phys. integrity index (lag)			2.721 (3.731)
Phys. int. lag squared			−4.745 (3.805)
Regional protest diffusion			0.881* (0.421)
Num.Obs.	524	524	464
AIC	668.0	646.6	294.9
BIC	19 685.0	685.0	381.9
F	3.698	4.782	2.204

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

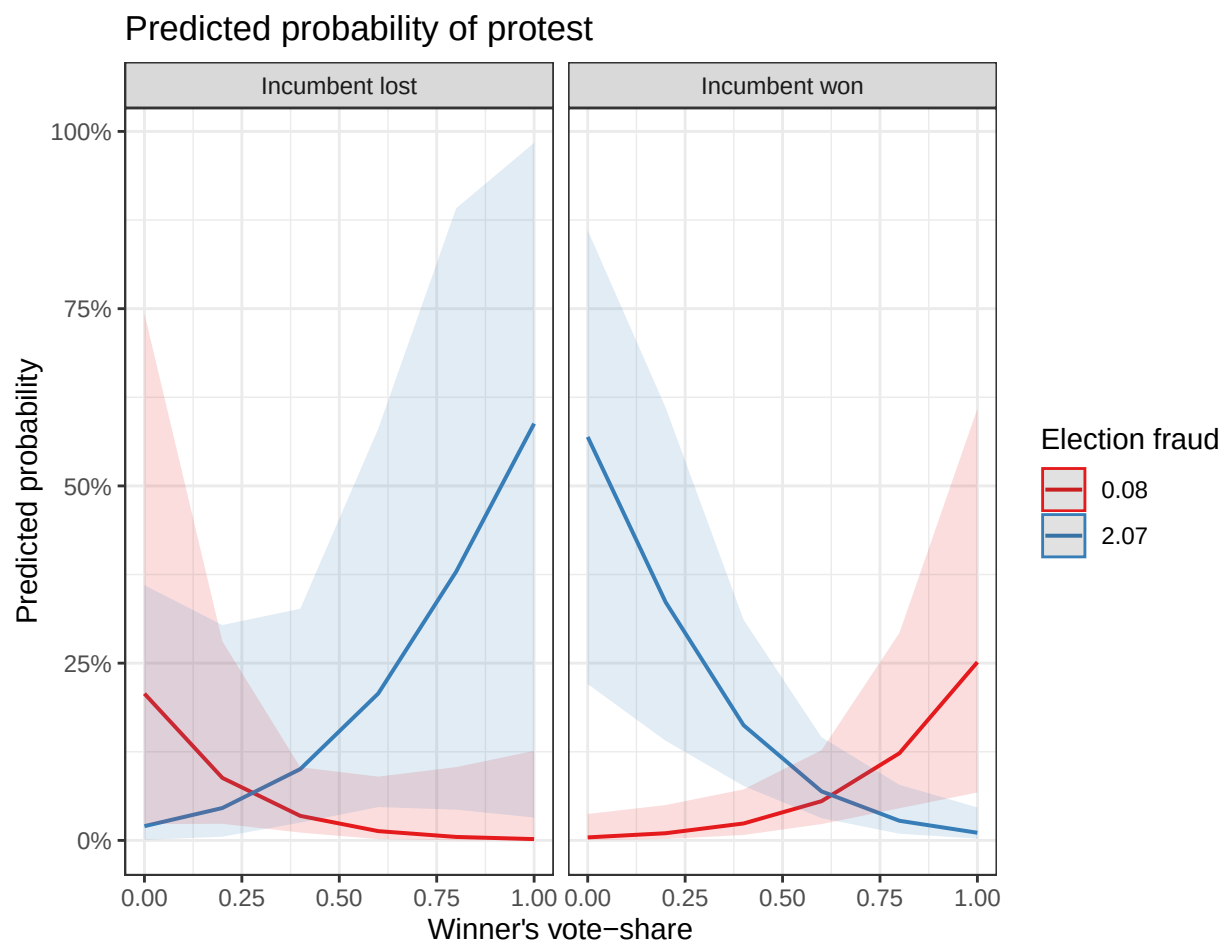


Figure 6: Fraud, winning vote-share, and probability of protest initiation

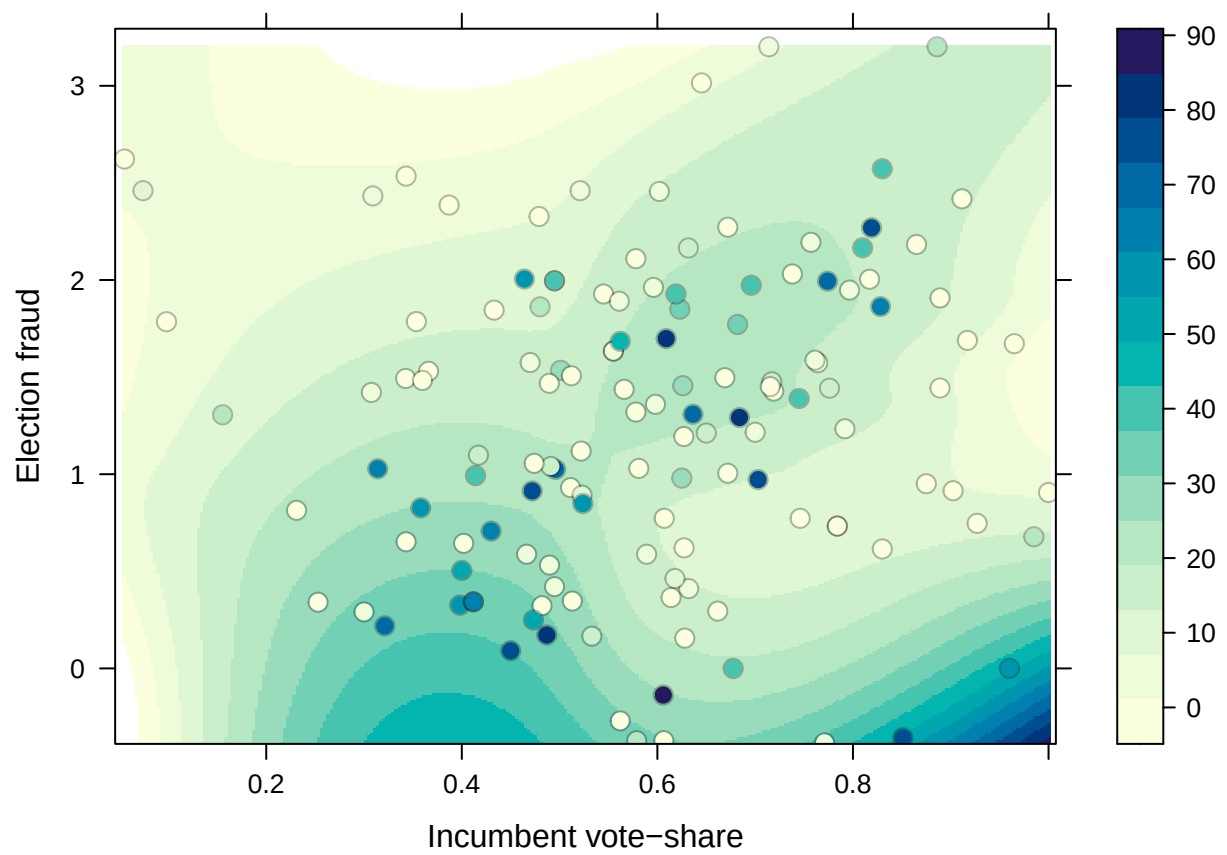


Figure 7: Fraud, incumbent vote-share, and duration of protest

protest movement; darker-colored dots indicate protest movements that survived longer. The pattern depicted in the plot is consistent with the regime-strength model of manipulation. First, there appears to be a zone of relative weakness along the bottom of the plot, where election fraud is minimal, across a wide range of vote-shares. Second, holding incumbent vote-share constant at a low value, incumbents appear stronger as we move up on the y-axis. Finally, there is an apparent zone of incumbent weakness running diagonally through the figure. This is consistent with the expectation that incumbents who produce moderate levels of fraud and middling vote-shares lack both the genuine popularity and the ‘control-based’ strength needed to quickly quell protest.

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